

Every strategy tested (YOLOv11n unless noted)

The winning ladder (adopted)		
Baseline (det, 640px)	0.668	
+ oversample + 1280px	0.749	
+ OBB task mode	0.759	
+ deg15 rotation	0.793	
+ blur-aug	0.799	no clean; +0.21 blur-test mAP@0.5
+ mixup — CV CHAMPION	0.804	
640px deploy (mixup+blur-aug)	0.757	current deployment recipe
Geometric augmentation — no gain beyond deg15		
Rotation 25°	0.781	worse than deg15
Rotation 30°	0.784	worse than deg15
Shear	0.779	worse than deg15
deg15 + shear 10°	0.788	no gain over deg15 alone
CPLID images restored to train	0.792	no gain over deg15
Lighter backbones — all below stock v11n/v8n		
Ghost graft (v11n / v8n)	0.68 / 0.62	vs stock 0.79 / 0.77
DWS graft (v11n / v8n)	0.70 / 0.69	vs stock 0.79 / 0.77
FasterNet/PConv graft (v11n / v8n)	0.72 / 0.70	vs stock 0.79 / 0.77
P2 stride-4 head (640px)	0.682	fresh-head init debt

External data — every route failed		
Community mixing (1:1–11.7:1)	0.58–0.65	DD AP; recall drop
Scale-matched curation	0.63–0.65	DD AP; recall drop
Pretrain-then-finetune (external)	below baseline	no run beat it
In-domain source pretraining	–7.5 mAP@0.5	genuine negative transfer
(control) in-distribution eval	≈0.91	DD AP; labels ARE learnable
Recall-focused levers @640 — none beat mixup alone (0.757)		
×6 defect oversample + mixup	0.747	worse
cls-loss gain 1.0 + mixup	0.748	flat
combo (×6 + cls-gain) + mixup	0.746	worse
Training-schedule ablations (Δ vs champion)		
Freeze 10 backbone layers	–0.051	hurts
Freeze 20 backbone layers	–0.299	collapses
300-epoch stage-1 (vs 150)	–0.005	overfits
Disable mosaic	–0.030	hurts
Remove scale-down aug	–0.023	hurts
Test-time augmentation (TTA)	+0.002	marginal, rejected
close_mosaic=20 @768	+0.001	DD +0.044, adopt-candidate